

Size evolution during laser-ignited single iron particle combustion

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Abstract

To further our understanding of iron particle combustion, especially in the liquid state, *in situ* optical measurements are performed for the size evolution and temperature of laser-ignited iron particles in a narrow size range around $45\text{--}55\mu\text{m}$. The particle size evolution is monitored by a high-speed shadowgraphy system, and the particle temperature is probed using a two-color pyrometer synchronized with the shadowgraphy system. Before a particle reaches the peak temperature, its diameter grows linearly with elapsed time. Near the peak temperature, three types of particle size behavior are detected, namely, smooth transition, micro-explosion, and rapid inflation. For particles with smooth transition, the size and temperature reach the maximum values at the same time, and thereafter the size remains constant, while the temperature drops until rapid solidification of the iron oxide droplet. The micro-explosion and rapid inflation clearly indicate towards a quick gas release inside the particle. The ambient oxygen concentration has a strong effect on the growth rate of the particle size before the peak temperature but no influence on the final particle size. The measured relative particle diameter at the peak temperature suggests that iron has been fully oxidized and the degree of oxidation could be higher than FeO. At the onset of solidification, gas release occurs again, resulting in a second inflation or burst of already inflated particles. This suggests that the oxygen dissolved in the liquid iron oxide is at least more than needed to form solid Fe_3O_4 , and the solidification process is accompanied with gaseous oxygen release via the phase transition: $\text{L2} \rightarrow \text{Fe}_3\text{O}_4(\text{s}) + \text{O}_2(\text{g})$ at 1855 K, where L2 represents liquid iron oxide with dissolved excess oxygen.

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1. Introduction

Iron powder is a promising energy carrier, which can potentially benefit the transition to a climate-neutral society, owing to its carbon-free nature, abundance, cyclability, high energy density, and low-volatile combustion at the normal conditions

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[1,2]. Towards practical applications of the iron fuel, experimental, theoretical, and numerical research has been carried out in recent decades, which advances our fundamental understanding on the combustion processes of iron particles and their suspensions. However, the state-of-the-art knowledge on iron combustion is still too low to support the development of a well-controlled iron-fueled energy conversion system.

The past experimental studies on iron combustion mainly focused on flame propagation in dispersed fine iron particles [3–6], where the importance of single particle burning was realized but only a little insight into the individual particle combustion behavior in the suspensions has been gained [3]. For example, Tang et al. [5] found that the particle combustion regime has a strong impact on the iron flame propagation. Due to the lack of fundamental characterization of fine iron particle combustion, numerical modeling attempts, seeking for better understanding of iron dust flames in ideally virtual environments, have been hardly proceeded except using a simplified reaction model [7]. The same challenge is also encountered in the development of metal dust flame propagation theories [8,9], where a compromise had to be accepted that particle combustion is described simply by its burn time.

To tackle one of the main bottlenecks for a deeper understanding of iron dust flames and future optimal utilization of the iron fuel, i.e., a physically accurate description of the single iron particle burning process, few *in situ* experiments focusing on isolated iron particle burning characteristics have been carried out very recently [10–13]. Ning et al. [11] measured the temperature histories of well-controlled single burning iron particles sized between 20–60 μ m in diameter. The results reveal that the combustion process of an iron particle undergoes several key steps: 1) ignition occurring in the solid-state, 2) iron melting, 3) intense exothermic reactions elevating particle temperature (fast burning), 4) particle cooling with a weak heat release rate (reactive quenching), and 5) solidification of the iron oxide droplet with an oxygen content close to that of Fe_3O_4 or higher. Based on the finding that the rate of solid-state iron oxidation is inversely proportional to the thickness of the oxide scale, a quantitative model has been achieved by Mi et al. [14] for the ignition and solid-state oxidization of fine iron particles. On the contrary, liquid-state combustion of iron is more intricately complex as discovered during combustion of thin iron rod in pressurized oxygen [15], and our current knowledge of the physics underlying liquid-state combustion of fine iron particles is still very limited. The measured durations of particle fast burning and reactive quenching is inversely proportional and insensitive to external oxygen concentration, respectively [10]. This suggests that the fast burning stage is probably governed by external oxygen diffusion,

while the reactive quenching step could be kinetic-controlled¹

The complexity of liquid-state iron particle combustion is also reflected by anomalous behavior observed during and after combustion such as micro-explosion [12,16], radiation jump [11], and hollow oxide particles [11,13,17]. Similar phenomena have also been observed during combustion of various metal particles, including Al [18], Ti [19], Zr [19–21], and their alloys [22–24]. By several dedicated experiments [20–22,25,26], it is consistently found that the occurrence of inflation, micro-explosion, and the spearpoint during combustion of Ti, Zr, and Al/Zr alloy particles is attributed to N_2 release. Although similar hypotheses have been proposed for iron [12,16], elaborate experiments are required to reveal more details and the underlying mechanism.

The present work is intended to *in situ* measure the particle size evolution during the entire burning process of a single laser-ignited iron particle based on high-speed shadowgraphy. This provides direct time-resolved combustion properties of iron particles. Besides, particle micro-explosion and formation of liquid iron oxide bubbles, which generate solid hollow oxide particle, are monitored with a large magnification. The adopted laser ignition technique offers an ideal condition for the investigation into the liquid-state combustion of iron particles by minimizing kinetically controlled oxidization in the solid-state at relatively low particle temperatures. Our study on the size evolution of burning iron particles will help to obtain further insights into the physics of liquid-state iron particle combustion and mechanisms of hollow oxide particle formation. The experimental data are also valuable for validation of iron particle combustion models.

2. Experimental methodology

2.1. Setup and optical diagnostics

Fig. 1 shows the schematic of the experimental setup for single particle combustion and *in situ* time-resolved optical measurements. The apparatus, except for the newly constructed shadowgraphy system, has been used previously to characterize the combustion of single iron particles [10,11], where a detailed description is available. Thus, only a brief summary is presented here for the already documented parts. Individual iron particles with a well-controlled velocity and trajectory are generated by an electrostatic-based particle generator. The generator features a coaxial flow configuration including a 0.45 mm inner diameter central jet for

¹ The kinetic here stands for not only chemical reactions but also internal ion diffusion with diffusivity described by an Arrhenius-like equation.

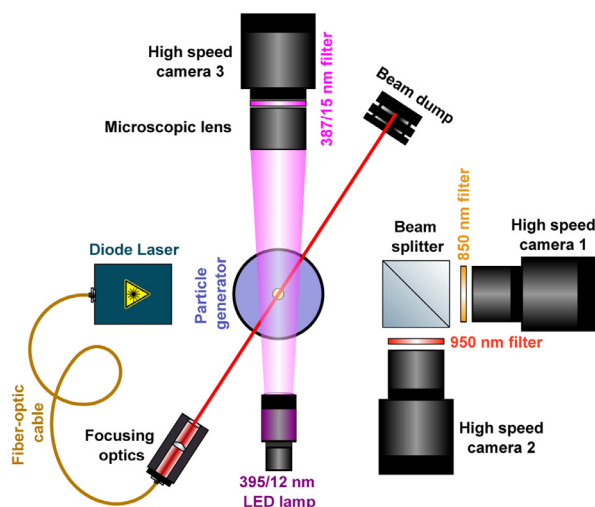


Fig. 1. Schematic of the experimental setup (top view). The high-speed cameras are synchronized and triggered by an external signal generator (not shown).

issuing aerosolized particles and a 20 mm inner diameter shroud flow for maintaining the particle velocity and a burning atmosphere. The velocities and compositions of the central jet and the shroud flow are regulated independently using 4 mass flow controllers. In the present experiment, the velocities of the central jet and the shroud flow are set at 43 cm/s and 50 cm/s, which provides a sufficient residence time within camera field of view for shadowgraphy. The gas compositions are between 14–21% volumetric O_2 diluted with N_2 and are kept equal for the central jet and the shroud flow. After generation, the particle moves upwards carried by the shroud flow and is ignited when passing the elliptical focal spot of a continuous 915 nm laser beam.

The thermal emissions of a burning iron particle filtered at 850 nm and 950 nm are recorded by 2 synchronized high-speed cameras (Photron SA3, model 60K-4GB), both equipped with an Irix 150mm f/2.8 macro lens. The high-speed cameras operate at a frame rate of 5000 fps. Then the surface temperature of the burning particle can be obtained using two-color pyrometry with a high time resolution for a wide temperature range. The calibration and measurement precision of the pyrometer have been extensively discussed in our previous work [11]. The particle size and behavior during the whole combustion process of the particle are visualized using a high-speed microscopic shadowgraphy system. A narrow-band LED lamp (Alite Lighting YT-1868UV) with a central wavelength of 395 nm and full width at half maximum (FWHM) of 12 nm is adopted as a light source for shadowgraphy. The lamp is modified by removing an internal pulse-modulation chip and then powered by a highly precise DC power supply (ELC-ALR3220), leading to a stable continuous light output. The

measured output power of the lamp is around 5 W. Another high-speed camera (Photron SA3, model 60K-8GB) is aligned with the lamp. This camera is time-synchronized with the other two cameras, i.e., a recording rate of 5000 fps is used. The shadowgraphy camera is equipped with a microscopic lens (Navitor zoom 6000), which achieves a resolution of 8.3 $\mu\text{m}/\text{pixel}$ and a field of view of 8.5 mm $\times$ 3.3 mm. A narrow-band-pass filter with a central wavelength of 385 nm and FWHM of 15 nm is installed in between the high-speed camera and the microscopic lens. The diffusive lamp light is concentrated to the microscopic lens using a built-in parabolic reflective cup and a condenser, which enables a shut-speed of 1/200,000 s for the shadowgraphy camera. Thereby, the contribution from the thermal radiation of the incandescent burning particle to the shadow image is suppressed to a level not exceeding the background noise. Due to large variation of particle temperature exceeding the measurement range of the pyrometer, only the camera installed with 850 nm filter is used simultaneously with the shadowgraphy system to indirectly indicate particle temperature evolution. The particle temperature is measured separately by increasing the shutter speed of the pyrometer to avoid signal saturation at particle peak temperature. At the same time, some information, approximately before particle melting, is lost.

2.2. Iron particle preparation

Iron particles with a similar diameter are prepared by sieving a commercial powder with a 99.8% purity from TLS Technik GmbH & Co. Fig. 2 shows the size distribution of the sieved particles, which is obtained by post processing images shot

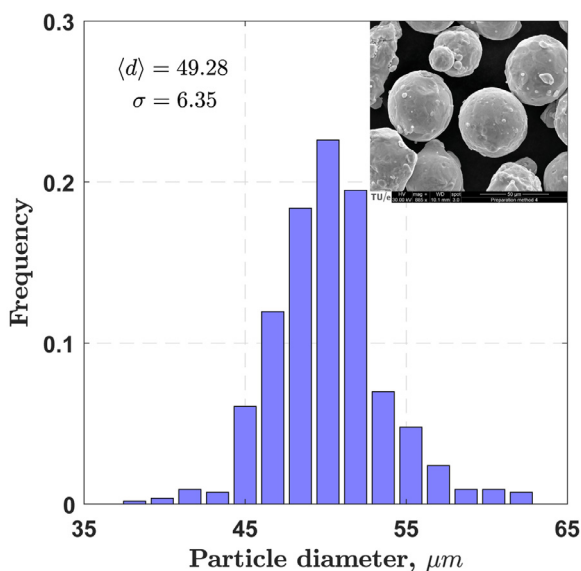


Fig. 2. Particle size distribution of the sieved iron powder. The arithmetic mean diameter $\langle d \rangle$ and standard deviation σ in μm are labeled. The inserted SEM image shows the particle morphology of the raw powder.

with a digital camera through a calibrated optical microscope. The particle size is mainly in the range of 45–55 μm . The calculated arithmetic mean diameter and standard deviation of the size distribution are 49.28 μm and 6.35 μm , respectively. The particle morphology is examined using a scanning electron microscope (SEM). As shown by the inserted SEM image in Fig. 2, the particles have a nearly spherical shape.

2.3. Data processing

The time-resolved 850 nm and 950 nm spectrum intensities of the thermal radiation from a single burning particle are retrieved from post-processing high-speed camera images using an in-house MATLAB code [10]. By calculating the ratio between the spectrum intensities, the particle surface temperature is obtained by two-color pyrometry [11].

As the wavelength of the light source for shadowgraphy is much smaller than the iron particle diameter, the incident light interacts with the particle geometrically. The light attenuation (LA) by the particle is proportional to the particle's cross section area. To obtain LA, the recorded shadow images are processed following the procedure below:

$$C = 255 \times \frac{A}{B}, \quad (1)$$

$$D = 255 - C, \quad (2)$$

where A, B, C and D represent a raw image containing a particle (particle image), a raw image excluding a particle (background image), an immediately processed 8-bit particle image with uniform

background (normalized image), and a finally inverted 8-bit particle image without background (inverted image). An example of the processing procedure showing raw and processed images is given in Fig. 3. The integrated pixel intensity over the bright areas of a particle in the inverted image is the needed LA. The integration process is the same as for retrieving the radiation intensity of a burning particle. The uncertainty induced by setting a threshold value for the integration in determining the relative particle size is estimated to be about 1%, because the intensity of pixels on the particle edge is relative small. When a particle is out of focus, more significant part of the shadow becomes out of the threshold for integration, which can induce a significant error. Therefore, obviously unfocused particles were discarded when processing the images. The validation of the proposed particle sizing approach is described in the supplementary material.

3. Results and discussion

3.1. Modes of particle size evolution

In the experiment, three types of particle size responses near the peak temperature, including smooth transition, micro-explosion, and rapid inflation, are observed. Based on the different responses, particle size evolution modes of 1) smooth transition (ST), 2) micro-explosion (ME), and 3) rapid inflation (RI) are defined. Fig. 4 shows typical radiation intensity trajectories filtered at 850 nm

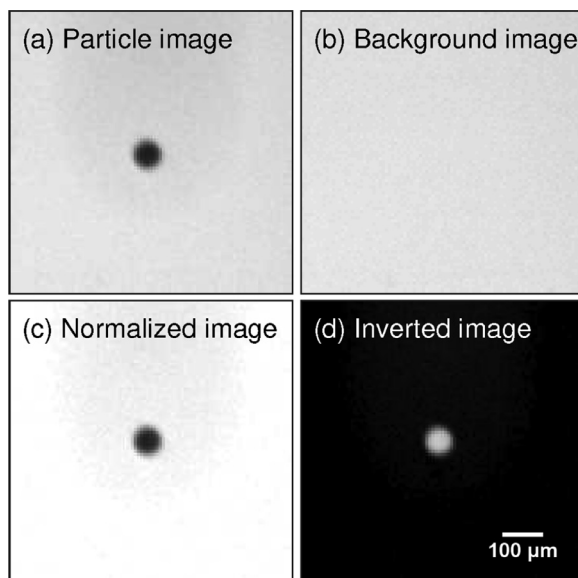


Fig. 3. Image processing procedure for obtaining the light attenuation by a particle. The particle image (a) and the background image (b) are raw photos with and without a particle recorded by the shadowgraphy high-speed camera, respectively.

and relative size evolution of particles burning in these three modes. The trajectories of radiation intensity shown in Fig. 4(a1)–(c1) are very similar except some small differences caused by particle size change near the peak temperature. The particle burning characteristics, such as combustion regime (diffusion-limited or kinetic-limited), and phase transitions, uncovered by analyzing the radiation intensity/temperature history have been reported in detail in our previous papers [10,11], and thus, are not repeated here. The radiation intensity trajectories are used to indicate the particle combustion progress, where particle peak temperature appears at the radiation intensity peak [11]. Fig. 4(a2)–(c2) show that the particle size increase nearly linearly during the fast burning stage until the proximity of the radiation intensity/temperature peak, as indicated by the linear fit with a high correlation coefficient ($R^2 > 0.9$). In addition, minor size shrinkage shortly after ignition is observed sometimes due to the melting of non-spherical particles. In order to reduce the uncertainty introduced by non-spherical particles to the mean relative particle size, the minimum particle diameter during the size evolution will be taken as d_0 in Section 3.2, which corresponds to the particle size right after melting for originally non-spherical particles. For ST, Fig. 4(a2) shows that, after the peak temperature, the particle size remains constant during the particle cooling phase within the time period of observation.

For ME, Fig. 4(b2) shows that at around 21 ms, about 2 ms before the peak temperature, the particle size surges within 0.5 ms and then collapses. At

the same time, a small spike in the radiation intensity profile can be seen due to the sudden change of the particle surface area. This suggests that the micro-explosion is caused by sudden internal gas release inflating the particle and breaking the “bubble” with a film of liquid iron oxides when the bubble size exceeds a threshold. Fig. 5(a) shows a series of snapshots of a particle within the instant of the micro-explosion. It is seen that at 15.4 ms the particle expands and at 15.6 ms the “bubble” bursts along with two small fragments ejected from the particle. The fragments move towards the lower-left side while the particle flies in the opposite direction, which means that the particle is pushed away from its original trajectory (vertically upwards) by gas ejection at one side on the particle surface. After the micro-explosion, the particle shape becomes elliptical and oscillates for about 2ms, and simultaneously fluctuations in the intensity profile are also observed, because the particle is returning to the spherical shape. Moreover, a nano-particle cloud is observed as shown by the mist around the particle in Fig. 5. Due to the micro-explosion, particles usually are pushed out of focus or the field of view of the camera. Therefore, Fig. 4(b2) does not show the data of particle size after 27ms.

For RI, Fig. 4(c2) shows that at approximately 17.5 ms, around 1.5 ms before the radiation intensity peak, the particle inflates rapidly and, hereafter, keeps increasing until the time of 19 ms when the size and the radiation intensity reach the peak at the same time. Unlike micro-explosion, the “bubble” does not break. From 19 ms, the particle cools

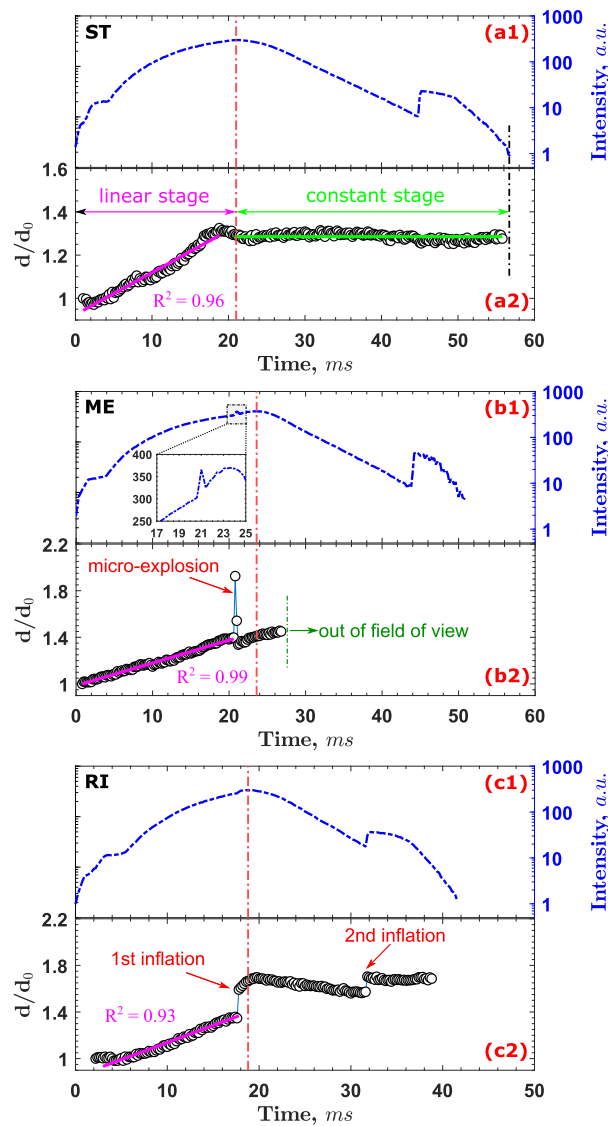


Fig. 4. Typical examples of radiation intensity (a1, b1, c1) and normalized particle diameter, d/d_0 (a2, b2, c2) as a function of time for particles burning in air in the size evolution mode of smooth combustion (a), micro-explosion (b), and inflation (c), respectively. The magenta line is a linear fit for the experimental data (black open circle) during the linear stage, and R^2 of the fit is > 0.9 . The green line is an average of the experimental data during the constant stage. The vertical red dash-dot line indicates the time when the particle reaches the maximum radiation intensity/temperature. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

down as indicated by the decreasing radiation intensity. At the same time, the particle shrinks due to the thermal shrinkage of gas trapped inside. At 32 ms, a second inflation occurs, and simultaneously a jump in the intensity profile is observed. The position of the radiation jump is conventionally referred to as the spear point [27]. The jump is mainly due to a rapid temperature increase triggered by sudden solidification of super-cooled iron oxide droplet [11]. The second inflation contributes

only minor to the radiation jump, which can be seen from the ratios between measured values of intensity just before and after the event. Note that the intensity is plotted on a logarithmic scale, and therefore a visually similar amplitude of jump in the intensity is much larger than a rise in the particle size. The fact that the second inflation and radiation jump are marginally correlated is also shown in Fig. 4(a), where the inflation does not occur but the radiation jump still appears. Unlike the decreasing

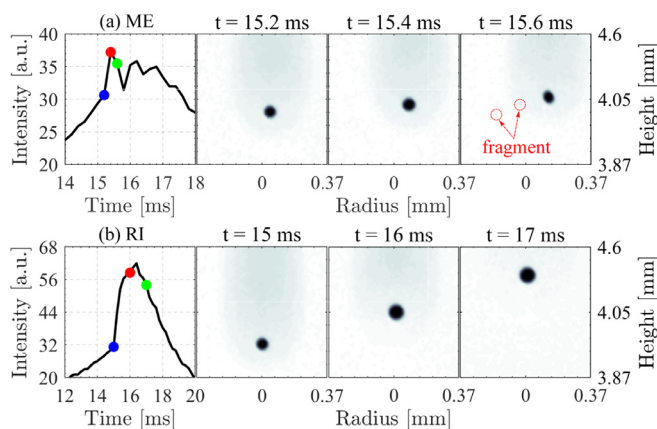


Fig. 5. Radiation intensity and snapshots of burning iron particles in air at three moments near the peak temperature for ME (top) and RI (bottom). The corresponding radiation intensity at the three moments are marked on the intensity profile by blue, red, and green dots. An artificial color scale is applied to the particle images to increase the visibility of the nano-particle cloud. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

particle size during cooling between 20 and -30 ms, after the second inflation, the measured particle size does not change because the particle quickly solidifies during the second inflation. The mechanism of the second inflation and rapid solidification will be discussed in detail in Section 3.3. The observed hollow burned particles in several experiments [11,13,17] are attributed to the inflation phenomena reported here.

Despite the seemingly unified reason of nitrogen release for the micro-explosion and inflation of multiple metals as aforementioned in Section 1, it is likely unjustified for iron. We conclude this from the recent experiment of Wright [28], as degassing of iron particles under high temperature vacuum conditions did not affect the micro-explosion of the iron particles burning in O_2/Ar . Instead, he found that the impurities significantly influence the micro-explosion. Similarly, the micro-explosion of burning steel particles generated by grinding shows a strong correlation with the carbon content [29]. This observation shows similarity with two recent experiments where the micro-explosion occurred when iron particles pass $CH_4/O_2/N_2$ sooting flames [12] and burn together with coal particles [16]. In short, the rapid gas release near the peak particle temperature during iron particle combustion, causing the micro-explosion or the first inflation, seems attributed to the impurity of carbon but the underlying mechanism remains unclear yet. The current observation could be a useful clue for further investigation.

3.2. Effect of oxygen concentration

Fig. 6 (a–c) shows that, at 14%, 17%, and 21% volumetric oxygen concentrations, the rela-

tive particle size evolution averaged for 10–15 particles with ST can be well characterized by a linear fit with $R^2 > 0.99$, and the slope is larger at a higher oxygen concentration, meaning that the particle reacts faster with external oxygen. At elevated oxygen concentrations, the nano-particle cloud is relatively dense, which influences the optical signals and therefore, no systematic measurements are conducted. At the end of the linear stage during particle size growth, as indicated by the vertical black dash-dot lines, the particle size reaches (or is very close to) its maximum value near the particle peak temperature (cf. Fig. 4). Fig. 6(d–f) shows the measured particle temperature histories and maximum equilibrium temperatures (near stoichiometry) predicated using Cantera [30]. The particle peak temperature decreases with decreasing O_2 concentration and is higher than the predicted equilibrium temperature. This is because iron particles burning in the diffusion regime are hotter than the equilibrium temperature at a fuel lean condition [7]. For the present single particle case, the equivalence ratio is practically zero.

The measured maximum relative particle size could be used to evaluate the degree of oxidation for the iron particle. To this end, the densities of liquid iron oxides at the present particle peak temperatures are first estimated based on available data at relatively low temperatures (see supplementary material). Fig. 7 shows the estimated relative particle sizes at the measured particle peak temperatures as a function of composition and the measured maximum relative particle sizes at the three O_2 concentrations. As the density of liquid iron oxide weakly depends on the composition, the estimated relative particle size only increases by around 6% from FeO to $FeO_{1.5}$. Although the uncertainty

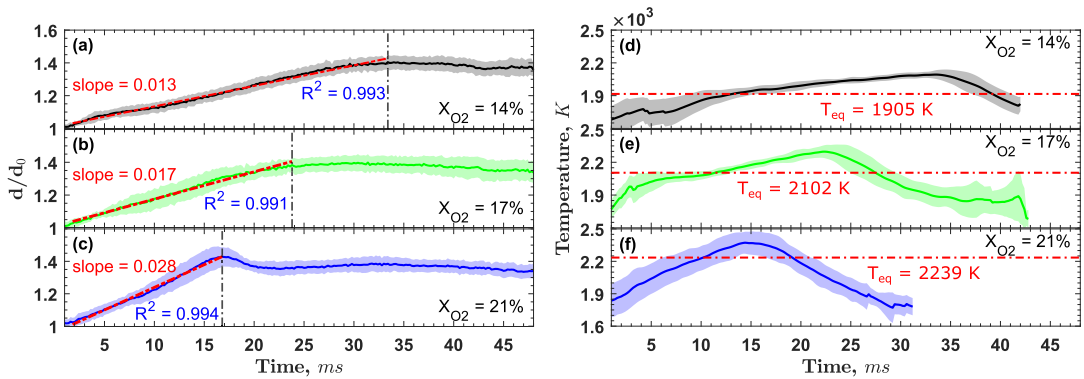


Fig. 6. Averaged relative particle diameter in mode ST (a–c) and temperature (d–f) as a function of time at oxygen concentrations of 14%, 17% and 21%. The solid lines are the mean values of the relative particle size or particle temperature calculated for 10–15 particles, and the shades represent the measurement uncertainty (standard deviation). For subfigures a–c, the vertical black dash-dot lines indicate the end of the linear stage of particle size evolution, and the red dash-dot lines are linear fits for the experimental data in the linear stage. For subfigures d–f, the horizontal red dash-dot lines indicate the maximum equilibrium temperature at the corresponding O_2 concentration.. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

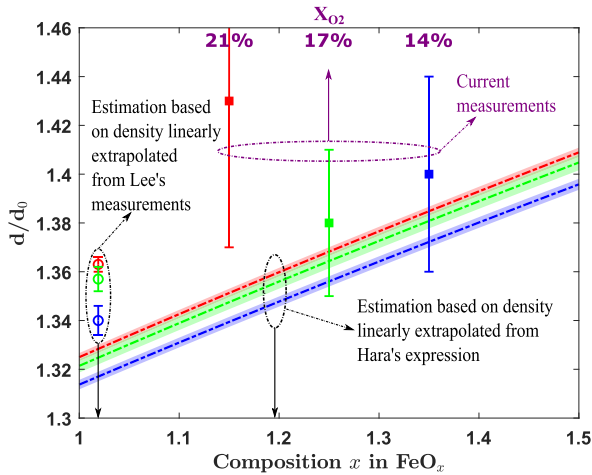


Fig. 7. Estimated relative particle sizes at the particle peak temperatures based on the density linearly extrapolated from measurements of Lee and Gaskell [31] and the expressions of Hara et al. [32], and comparison with the current measurements. The red, green and blue colors correspond to measured particle peak temperatures of 2390 ± 43 K, 2297 ± 92 K, and 2093 ± 52 K at 21%, 17%, and 14% O_2 , respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of the present measurement is already small, about 4%, a good estimation of the composition is still hard to be made. The extrapolation due to inadequate data on liquid iron oxide density is one of the principal uncertainties involved. The average values of the measured relative size suggest that the oxygen composition of the particle is probably higher than FeO at the peak temperature. A more prospective identification of the oxidation state could be achieved by quenching burning particles and analyzing the composition of the collected products. Besides, nano oxide cloud is observed

around a particle for 17% and 21% O_2 cases, and the cloud density decreases with decreasing O_2 concentration. Specifically, nano-particle cloud forms surrounding the particle and is gradually elongated due to the slip velocity between the particle and gas. Very close to the peak temperature, the nanoparticle could detach the burning particle, suggesting that nanoparticle formation stops. Since the vapor pressure from Fe(l) evaporation is more than one order of magnitude higher than that from the dissociation of FeO(l) and Fe_3O_4 (l), and it increases exponentially with increasing temperature [11], the

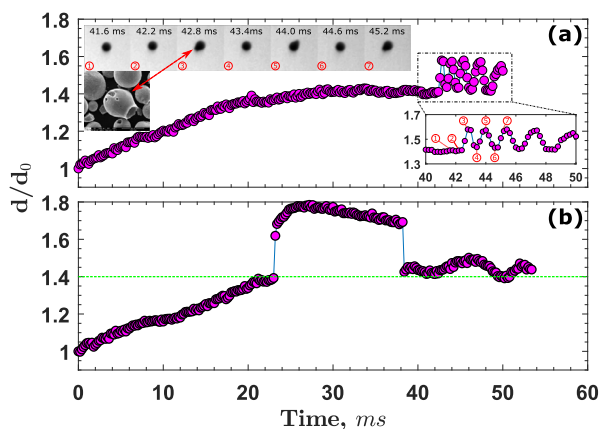


Fig. 8. Relative size evolution of burning iron particles with (a) only the “second” inflation and (b) with the first inflation and collapse at the spear point. The oxygen concentration is 17% for both scenarios.

end of the nano-particle formation also suggests that liquid iron has been just fully consumed.

Based on the extrapolated densities of liquid iron oxide, the effect of thermal shrinkage on particle size change is evaluated. The result shows that the particle size only decreases by 3% (using the density extrapolated from data of Lee and Gaskell [31]) or 5% (using the density extrapolated from expressions of Hara et al. [32]) when cooling down from 2400 K to 1880 K (the particle temperature just after the radiation jump shown in Fig. 4 [11]). Therefore, the thermal shrinkage effect on particle size during the cooling stage in the experiment is negligible. The observed constant stage of size evolution suggests that further oxidation could be minor after the peak temperature.

3.3. Mechanism of the second inflation

Our previous study [11] showed that, at the spear point, a rapid phase transition of super-cooled iron oxide droplets occurs, which triggers an abrupt particle temperature increase and subsequently the radiation jump. On the basis of the measured particle temperature right after the brightness jump (1880 ± 70 K), the phase transition is identified as (1) $L2 \rightarrow Fe_3O_4(s)$ at 1869 K or (2) $L2 \rightarrow Fe_3O_4(s) + O_2(g)$ at 1855 K, where L2 represents liquid iron oxide. If the dissolved oxygen in the liquid iron oxide droplet is more than that needed to form stoichiometric $Fe_3O_4(s)$, gaseous oxygen can be released from the particle via phase transition (2). In the experiment of the burning thin iron rods performed by Muller et al. [15], also a gas release process was observed during cooling until complete solidification of iron oxide, which could be attributed to phase transition (2). Therefore, the second inflation (cf. Fig. 4(c2)) at the spear point probably results from fast releasing oxygen generated during rapid solidification of super-cooled

iron oxide droplets. Besides, the second inflation is partially attributed to the thermal expansion of the trapped gas that is released near the peak temperature and triggers the first inflation or micro-explosion. Nevertheless, our estimation indicates that the gas thermal expansion only is insufficient to inflate the particle to the observed size.

Fig. 8 presents an event of internal gas release and simultaneous abrupt solidification. As seen in Fig. 8(a), the first inflation does not appear near the end of the linear stage of particle growth (around particle peak temperature), and only the “second” inflation occurs at the spear point after the constant stage where the particle cools down (unlike particles with trapped gas showing decreasing particle size at this stage). This demonstrates that the second inflation cannot be solely caused by the thermal expansion of trapped gas. After the “second” inflation, the measured value of relative particle size fluctuates periodically as can be seen in the enlarged view. This fluctuation is due to the particle becoming non-spherical and rotating, as can be seen from the inserted snapshots in Fig. 8(a), leading to a varying particle cross section projected to the camera. At ③ 42.8 ms, ⑤ 44.0 ms, and ⑦ 45.2 ms, the particle shapes resemble and the measured particle sizes are almost identical, which suggests that the particle has already solidified, otherwise the particle should have been contracted to a spherical shape by surface tension. Right before the inflation at 42.4 ms, the particle is in the liquid state suggested by the unchanged spherical shape shown in the snapshots at ① 41.6 ms and ② 42.2 ms. The observed phenomenon can be explained by that, at the spear point, quick gas release inflates the liquid iron oxide droplet, resulting in a “bubble” that sometimes bursts when the amount of released gas is sufficient. At the same time, rapid solidification (if fast enough) freezes the moment of “bubble” burst. In the captured combustion products, tulip-shape par-

ticles, resembling that *in situ* observed (e.g., at 42.8 ms in Fig. 8(a)), can be found as shown in the inserted SEM image. The particle in the SEM image shows a clear sign of gas ejection from the inside of the particle. It is observed that the second inflation mainly occurs after the first one, and the occurrence of the second inflation alone is rare (< 5%). This can be explained by the fact that much higher initial pressure has to be built up to generate a new bubble than to inflate an the existing one.

Besides the second inflation, it is also observed that the “bubble” already inflated near the peak particle temperature can collapse due to the gas release at the spear point. Fig. 8(b) shows a typical example of this scenario. At 38.4_ms, the measured relative particle size is close to that right before the first inflation, which means that the trapped gas is quickly released due to accelerating particle contraction by surface tension. After the burst, the measured value of relative particle size also shows periodical fluctuations similar to the particle in Fig. 8(a), meaning that the droplet solidifies before it could fully contract to a spherical shape. The simultaneous occurrences of rapid gas release and solidification at the spear point, together with the previously measured local particle temperature, leads to the conclusion that fast phase transition of super-cooled iron oxide droplets, i.e., $L2 \rightarrow Fe_3O_4(s) + O_2(g)$, happens. The oxygen-to-iron ratio in L2 is at least higher than 1.33 ($Fe_3O_4(s)$), and possibly exceeds 1.5 ($Fe_2O_3(s)$) as found by Steinberg et al.[33]. Since the degree of super-cooling may vary from particle to particle, the oxygen release rate could also change, leading to different particle size behavior at the spear point, i.e., inflation, bust, and no change.

4. Conclusions

In situ optical measurements of particle size evolution and temperature are performed based on high-speed shadowgraphy and two-color pyrometry, respectively. The measured particle size increases linearly with time until around the peak temperature. The growth rate of particle size in the linear stage is strongly influenced by external oxygen concentration. At the peak particle temperature, the end of nano-particle formation indicates the depletion of iron, and the estimated relative particle sizes based on extrapolated densities of liquid iron oxide suggest that the degree of oxidation of iron particles could be higher than FeO . Three particle responses are observed near the peak temperature, i.e., smooth transition, micro-explosion, and rapid inflation. For smooth transition, the particle size barely changes after the peak temperature, whereas for inflated particles, their sizes decrease due to thermal shrinkage of trapped gases and particle cooling. The second inflation and permanent deformation of particles are discovered at the spear

point, suggesting that $O_2(g)$ release and rapid solidification occur simultaneously owing to the phase transition: $L2 \rightarrow Fe_3O_4(s) + O_2(g)$ of super-cooled iron oxide droplets, where L2 represents liquid iron oxide with dissolved oxygen exceeding that required for stoichiometric solid Fe_3O_4 .

Declaration of Competing Interest

There is no conflict of interest for this work.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.proci.2022.07.030](https://doi.org/10.1016/j.proci.2022.07.030).

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